Supporting Information

CoCo — A Low-Cost Open Electrode Coating Platform
with Integrated Process Characterization

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Bill of Materials

Everything you need is available from standard laboratory or consumer electronics suppliers. Prices are approximate costs at time of purchase in Germany.

Category	Component	Qty	Cost (€)	Notes
Base platform	Creality Ender-3 **SE**	1	169.00	Standard retail (see note below)
3D-printed	Blade holder	1	–	PLA, see repository
	Vacuum plate	1	–	PLA, see repository
	Blade storage rack	1	–	PLA, see repository
	Ejection tray	1	–	PLA, see repository
	Rail mount	1	–	PLA, see repository
	Plate clamp corners	4	–	PLA, see repository
	PLA filament total	∼200 g	10.00	Standard PLA
Consumables	Utility blades 18 mm	1 pack (10)	2.87	
Vacuum	Venturi injector	1	6.50	
	Pneumatic tubing 4 mm	∼1 m	1.56	
Fasteners	M3×14 mm screws	8	–	Blade Storage
	M3×18 mm screws	2	–	Blade holder top
	M3×6 mm screws	2	–	Blade holder side
	M3×3 mm screw	1	–	Blade holder right
	M3 washers (OD 5 mm, t 1 mm)	3	–	Blade holder
	M4×40 mm screws	4	–	Bed spring mounts
	Bed levelling spring kit	1 set	∼6.00	e.g. YIXISI improved bed springs (Amazon)
	Fasteners total	–	10.00	
Magnets	Neodymium magnets 2.7 mm×6 mm	2	<1.00	
Total			**≈200**	

The Ender-3 SE model is required. The visually similar Ender-3 KE lacks a serial communication interface and is incompatible with CoCo.

3D-Printed Components

All custom parts were designed in Fusion 360 (Autodesk Inc.) and printed in PLA with a Bambu Lab X1C. STL, STEP, and F3D files are available at https://github.com/Helge-Stein-Group/coco_coater (MIT Licence).

Part	Setting	Value
All parts	Material	PLA
All parts	Build plate	Smooth PEI
All parts	Support	Tree support
All parts (except Vacuum Plate)	Profile	0.2 mm Standard (Bambu Lab X1C)
Vacuum plate	Profile	0.08 mm Extra Fine (Bambu Lab X1C)

Blade holder orientation: Print on its side with the two screw holes facing upward and use tree supports at standard resolution.

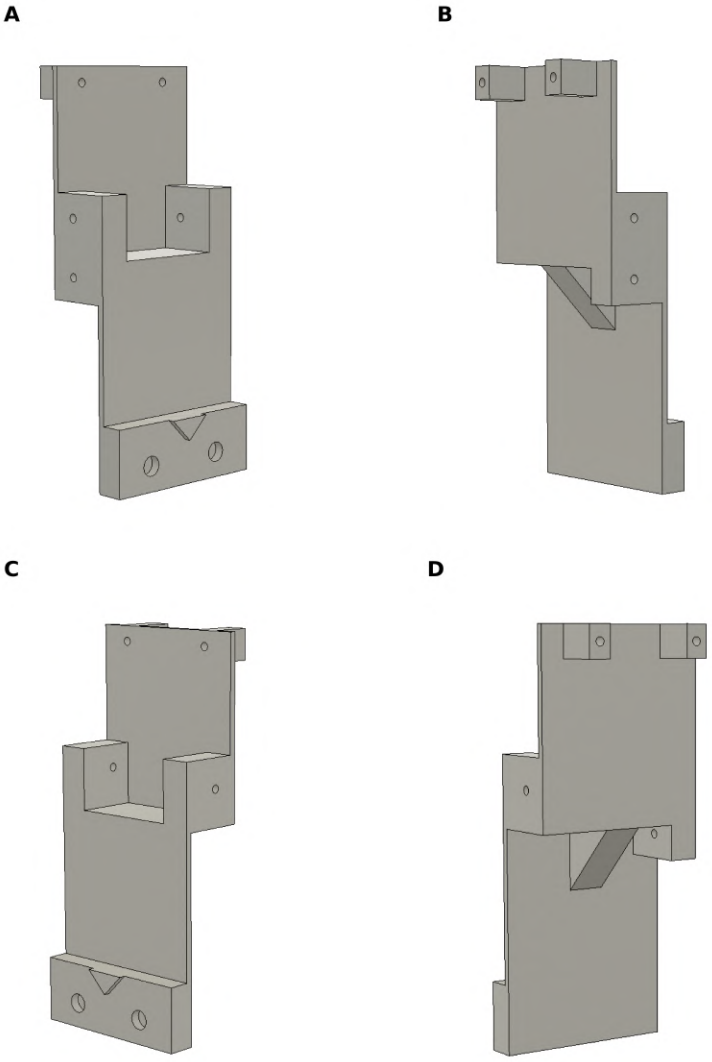
Vacuum plate orientation: Print face-down (top surface against build plate) at extra-fine resolution. The build-plate side of a PLA print is smoother than the top surface, which matters because this surface contacts the substrate foil directly during coating.

For organic solvents (e.g. NMP), verify PLA compatibility or replace affected parts with PETG, which offers superior chemical resistance. PLA is fully compatible with the water-based slurry systems used in this work.

To Print

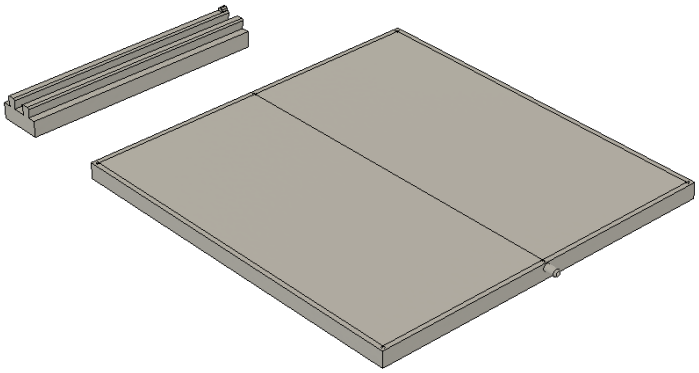
1x Blade Holder

The **blade holder** (below) attaches to the print-head carriage. The two circular recesses at the front are made for neodymium magnets that hold the coating blade. The triangular stopper above the recesses ensures the blade always extends below the holder and self-aligns parallel to the substrate when it descends.



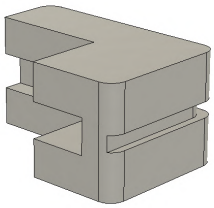
1x Vacuum Plate

The **vacuum plate** distributes suction pressure through a perimeter groove and central channel. A short rail segment is glued to its rear left edge, this serves as the docking point for the blade storage station. The vacuum plate and the rail cannot be printed as one piece because the vacuum plate is printed face-down for surface quality reasons.



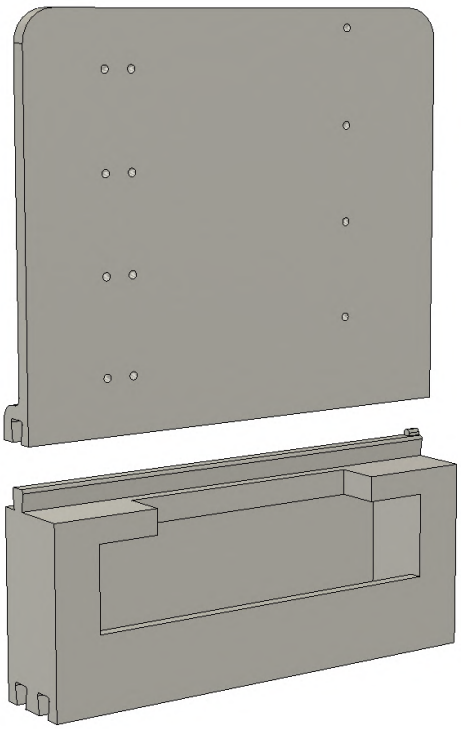
4x Corner Clamps

The **corner clamps** lock the vacuum plate against the printer bed. Each clamp has a notch that locates onto the bed frame; a cord routed along the outer groove of all four clamps is tightened to clamp the plate firmly in place.



1x Blade Storage and Ejection Tray

The **blade storage rack and ejection tray** slide onto the bonded rail segment. The rack holds up to four utility blades in numbered slots. The tray collects used blades automatically after each coating run.



Assembly Instructions

What you need: hex key set, small files (for thread removal), cyanoacrylate adhesive. Estimated time: ≈8 hours.

The Creality Ender-3 SE provides three motion axes: Y (print bed, coating direction), X (print head, blade width direction), and Z (blade height). The original extruder assembly is removed during assembly and replaced with the custom blade holder.

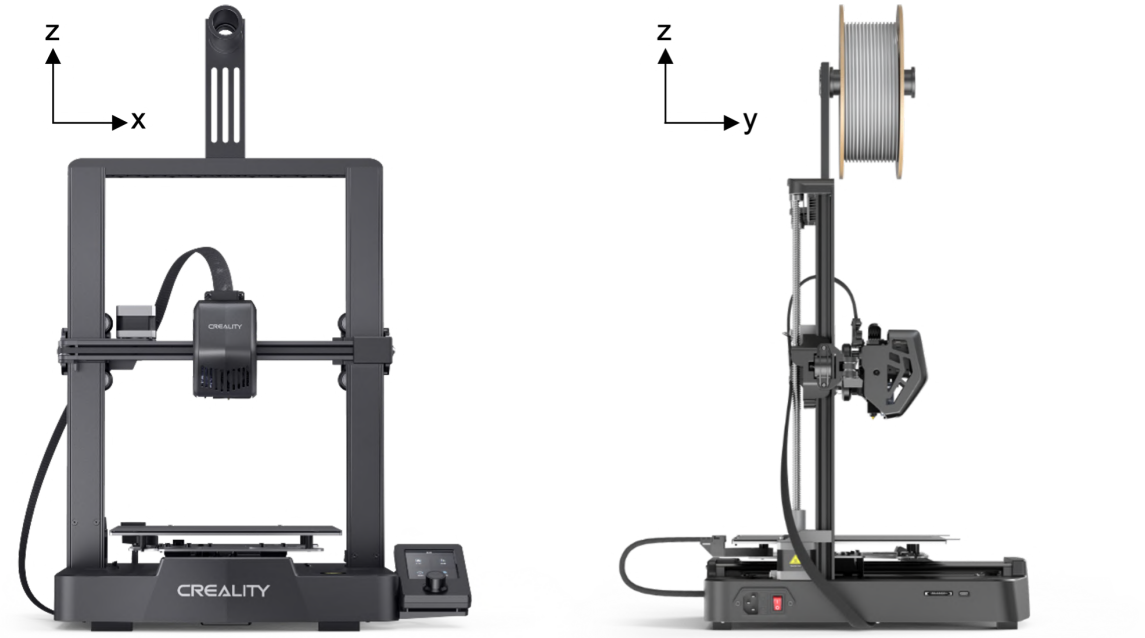


Image source: Creality (https://store.creality.com/de/products/ender-3-v3-se-3d-drucker)

STEP 1 — Bed Modification

The four rigid spacers that fix the bed height are replaced with spring-loaded mounts so the vacuum plate can be levelled manually at each corner.

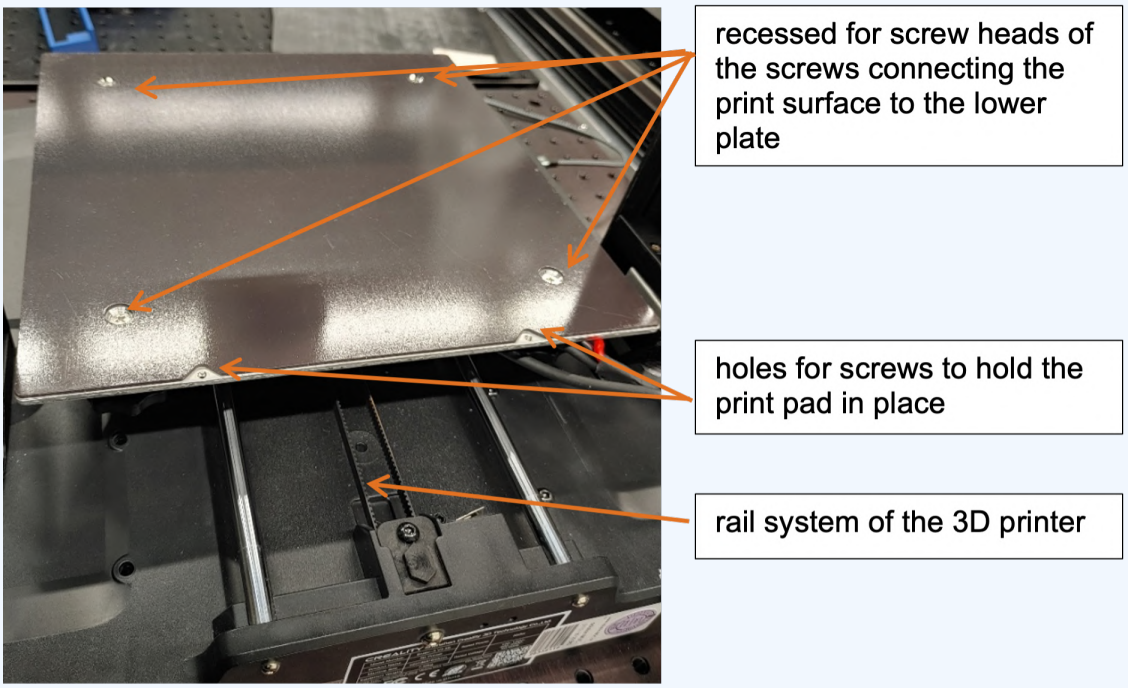
1. Remove the print pad to expose the four M4 screws that connect the print bed to the lower plate.



Image source: Creality

(https://store.creality.com/de/products/ender-3-v3-se-3d-drucker)

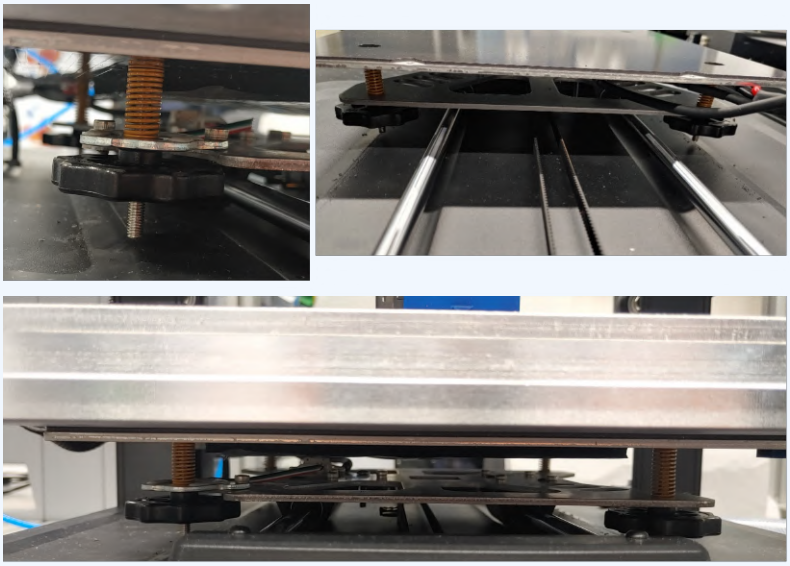
2. Unscrew and remove all four M4 screws and rigid spacers.



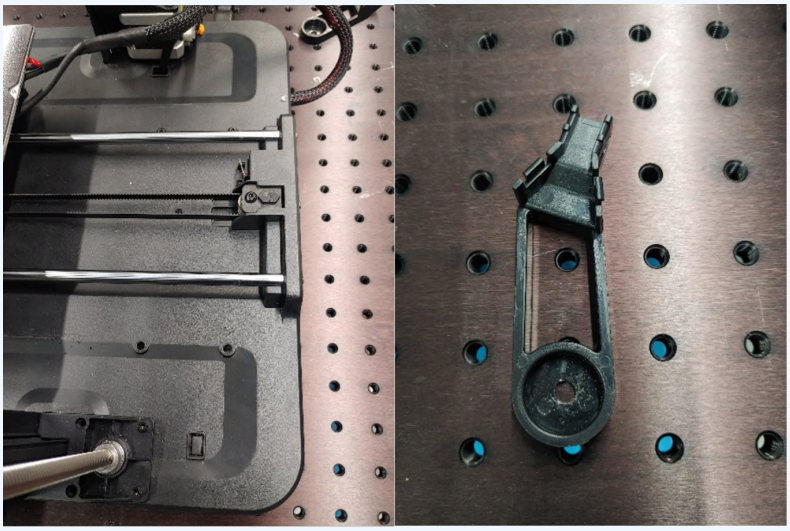
3. Using a small file or drill bit, open up the four screw holes in the lower plate so the screws can slide freely through without catching on a thread.
4. Insert the springs, thread the M4×40 mm screws from above, and hand-tighten the adjustment knobs from below. Set the bed to approximately mid-spring travel.



Image source: Amazon (https://www.amazon.com.be/-/en/YIXISI-Printer-Leveling-Spring-Screws/dp/B09KKRQ7WH)



5. Remove the two screws at the rear right of the black printer base. The longer replacement screws will scrape against them at higher Y-positions and cause vibration during coating if left in place.
6. Remove the cable management bracket between the print bed and the lower plate. It causes the bed to flex when the corner knobs are adjusted.



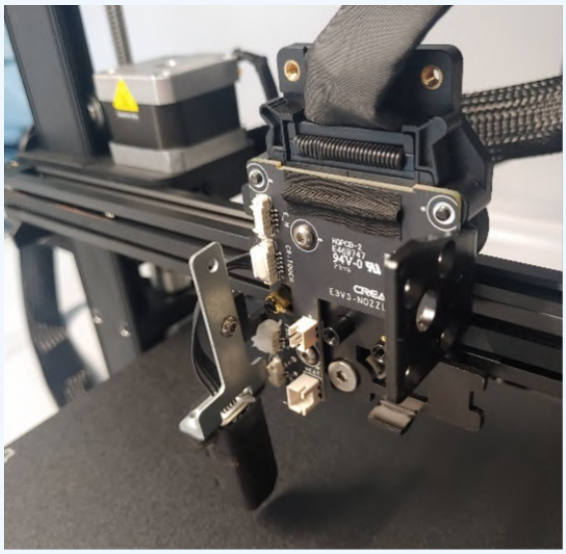
7. Remove the two guide screws at the rear edge of the print bed. They were used only for print-pad alignment and protrude above the surface, interfering with the vacuum plate.



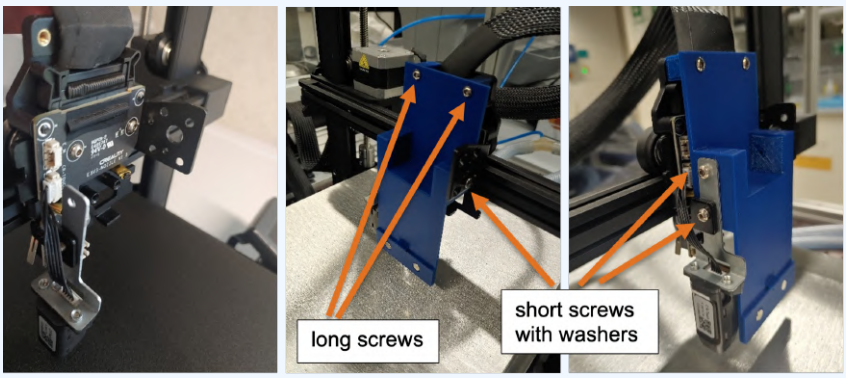
Image source: Creality
(https://store.creality.com/de/products/ender-3-v3-se-3d-drucker)

STEP 2 — Extruder Removal and Blade Holder Installation

1. Disconnect the stepper motor connector and remove the M3 mounting screws to detach the extruder assembly from the print-head carriage. Remove the hotend and all heating elements. **Keep the CR Touch sensor**: it is retained as the Z-axis end-stop and is integral to the blade holder design.



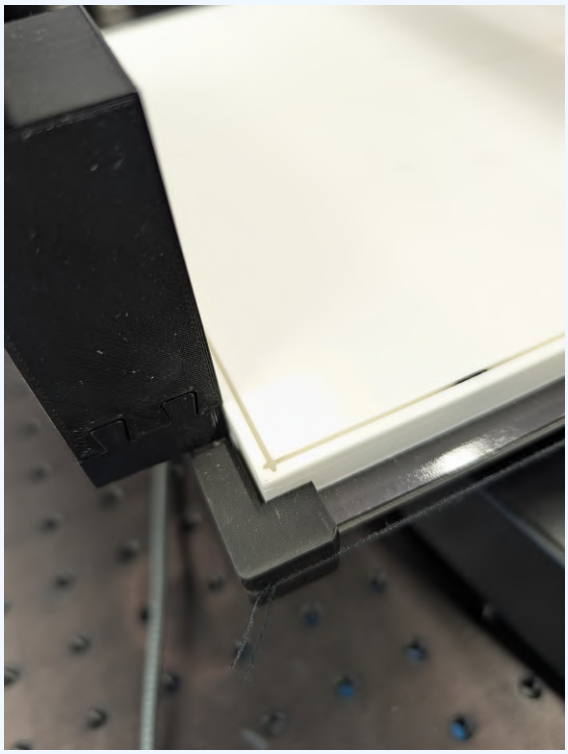
2. Mount the blade holder to the carriage using the vacated screw holes:
   - Top two holes: M3×18 mm screws
   - Left side (CR Touch): M3×6 mm screws + M3 washers (OD 5 mm, thickness 1 mm)
   - Right side: M3×3 mm screw + washer



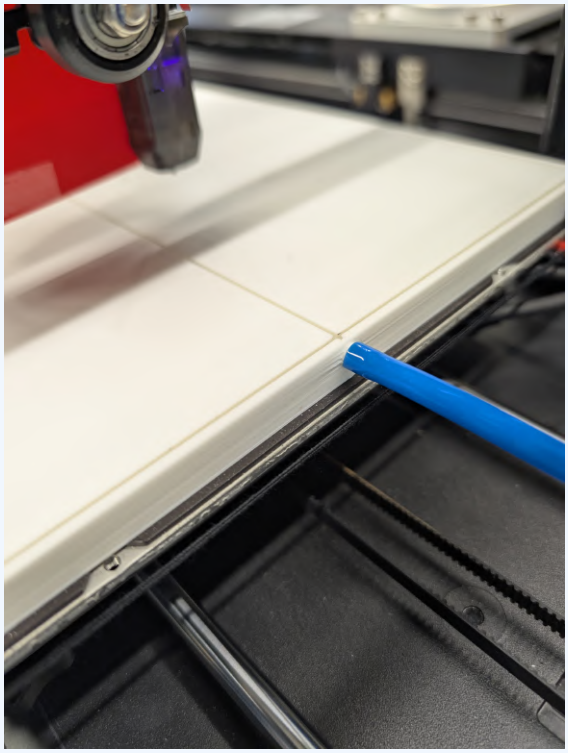
3. Press two neodymium magnets (2.7 mm, height 6 mm) into the two recesses on the front of the blade holder. They should sit tight.

STEP 3 — Vacuum Plate Installation

1. Set the vacuum plate onto the printer bed after removing the print pad.
2. Place the four corner clamps, one per corner. Slide each clamp so its notch locates onto the vacuum bed frame, then route a cord along the outer groove of all four clamps and tighten to clamp the plate firmly against the bed.



3. Connect the Venturi injector to the laboratory compressed-air supply using 4 mm pneumatic tubing. Connect the injector outlet to the vacuum port on the side of the vacuum plate.



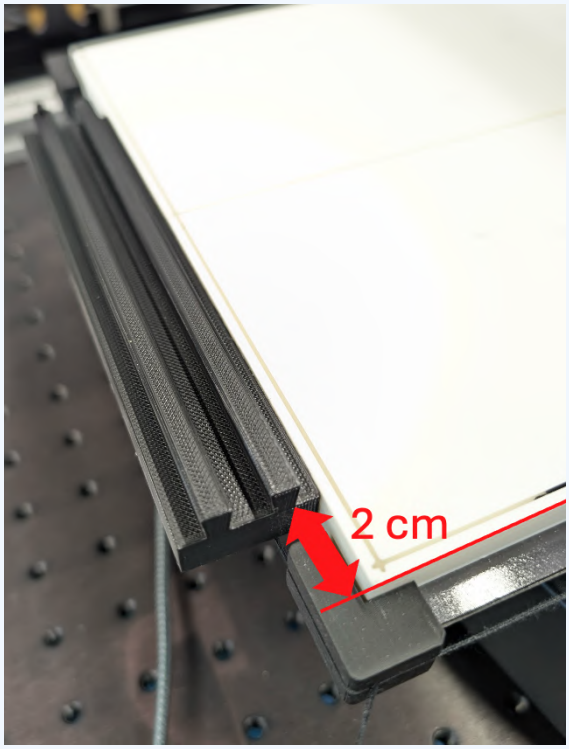
4. Verify suction by holding a sheet of paper over the plate — it should stay firmly in place without slipping.

The Venturi injector operates at standard laboratory pressure (typically 4–6 bar). Check all pneumatic connections before each session and replace any damaged tubing immediately.

STEP 4 — Rail System

The rail must be positioned exactly 2 cm from the corner edge of the vacuum plate. This offset accounts for the space occupied by the corner clamps and ensures the rack sits within the programmed X-range of the blade pickup movement. Incorrect positioning will interfere with the programmed motion sequence.

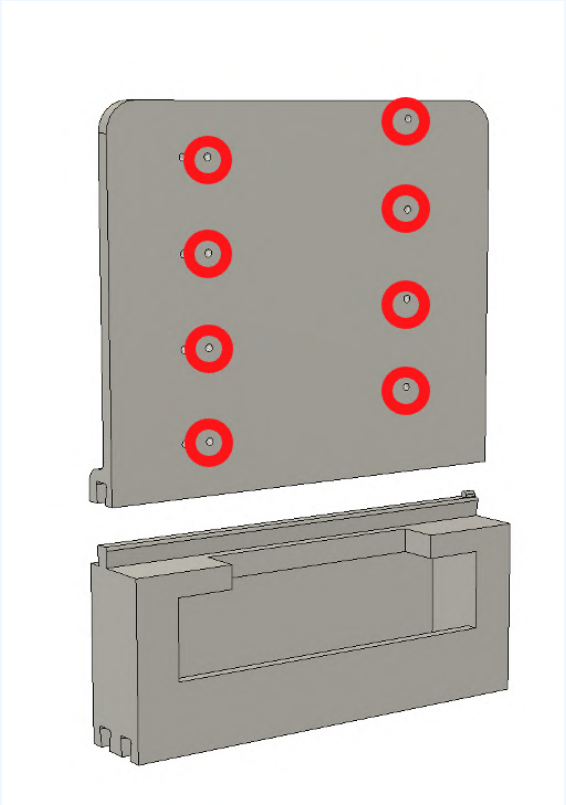
1. Clean and dry the bonding surface on the rear left edge of the vacuum plate.
2. Measure 2 cm from the corner edge and mark the position.



3. Apply cyanoacrylate adhesive on the rail edge and press the rail segment firmly in place. Allow to cure fully before proceeding.

STEP 5 — Blade Storage Station

1. Insert four M3×14 mm screws into the right-side holes of the storage station.
2. Insert four M3×14 mm screws into the inner left-side holes of the storage station. The outer holes are provided for compatibility with longer blades if needed.
   The screws form the blade retention pegs on which the blades rest in the rack.



3. Slide the blade storage rack and ejection tray onto the bonded rail.
4. Load up to four utility blades into the slots (1–4). You can number them if you like.



Each slot corresponds to one coating field: blade 1 for Field 1, blade 2 for Field 2, and so on. The ejection tray below the rack collects used blades automatically after each run.

Although blade pickup and ejection are automated, handle blades carefully when loading the rack or retrieving from the tray. Wear cut-resistant gloves.

STEP 6 — Software Installation

1. Install Python 3.8 or later.
2. Open a terminal and install the required dependencies:

```
pip install customtkinter pyserial numpy openpyxl matplotlib
```

3. Connect the printer to the computer via USB.
4. Identify the serial port:
   - **Windows:** Device Manager → Ports (COM & LPT)
   - **Linux** / **macOS:** run `ls /dev/tty*` before and after plugging in the printer — the new entry is the correct port
5. Open `coco_gui.py` and update the port string, then launch the GUI:

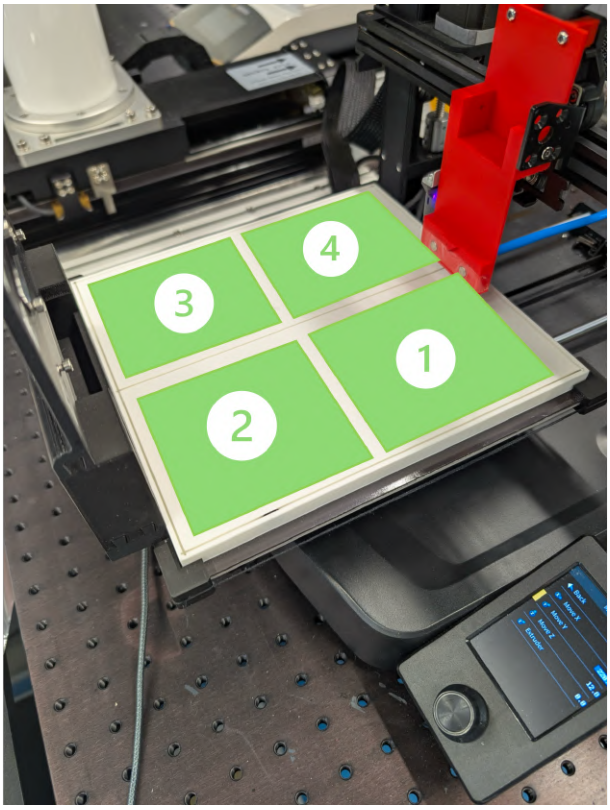
```
python coco_gui.py
```

6. Check that the status bar shows a successful connection at 115200 baud.

CoCo requires the Ender-3 SE. The Ender-3 KE is visually identical but lacks a serial interface and cannot be used.

Coating Field Layout

The vacuum plate provides four independent coating fields of 8×6.5 cm each.

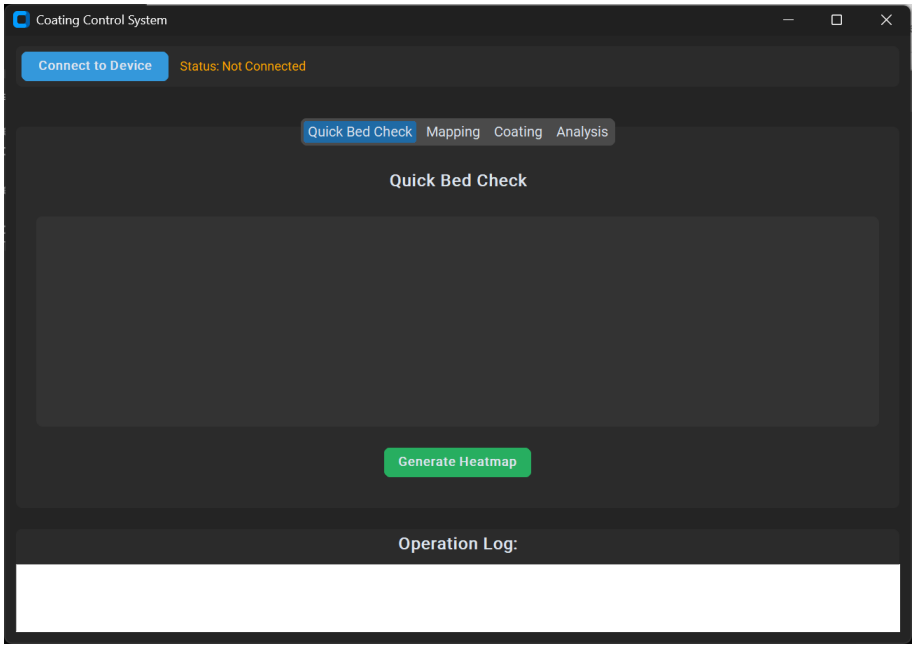


GUI Operation

The GUI guides the user through four sequential tabs. Open the GUI by running `python coco_gui.py`.

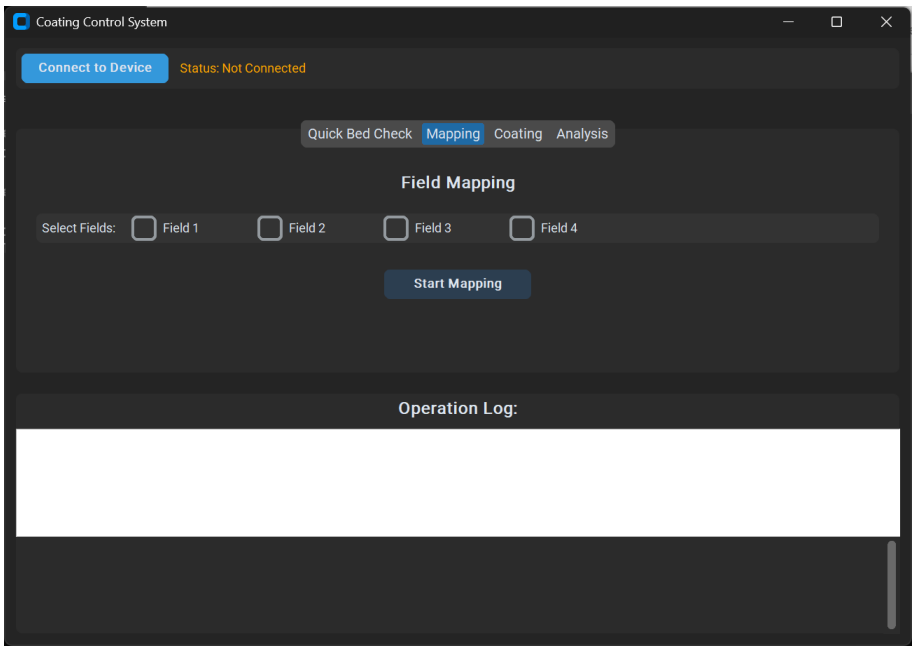
Tab 1 — Quick Bed Check

Use this tab for initial setup and every time the vacuum plate is reinstalled. Click *Generate Heatmap* to run a coarse bed scan using the Marlin bilinear levelling routine. The result is displayed as a height map divided into the four coating quadrants. If the peak-to-valley deviation exceeds 200 µm, adjust the spring-loaded corner knobs and scan again.



Tab 2 — Mapping

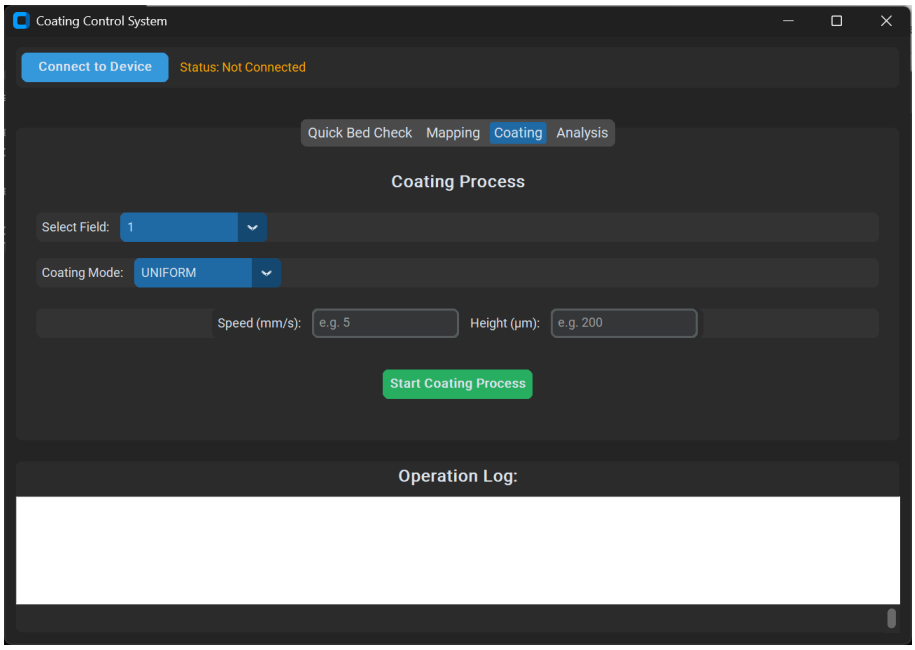
Select one or more fields with the checkboxes and click *Start Mapping*. The probe traverses the substrate on a 7 mm grid and saves the height map automatically. This mapping step must be completed before every coating run because the coating routine uses the map for row-wise blade height correction.



Tab 3 — Coating

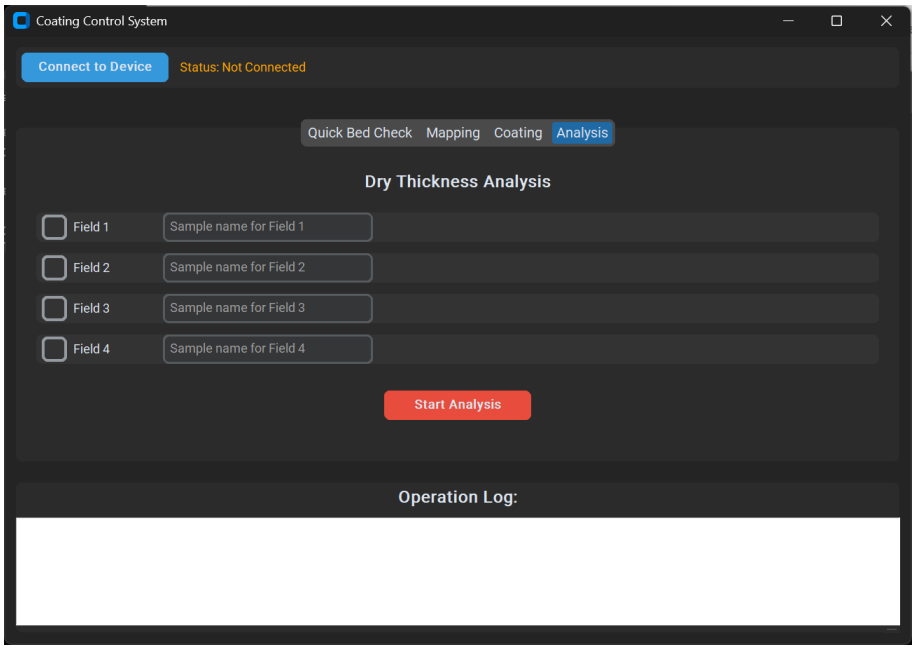
Select the field and deposition mode, enter the parameters, and click *Start Coating Process*. The GUI then prompts you to apply slurry before the blade begins moving.

- **UNIFORM** — one gap height and one speed for the whole field
- **ZONE** — independent gap heights and speeds for Zone 1 and Zone 2
- **GRADIENT** — one speed, start gap height, and end gap height; the gap changes linearly across the field



Tab 4 — Analysis

Select the fields to analyse, enter a sample name for each, and click *Start Analysis*. The system re-maps the coated surface, computes the dry thickness at every grid point, and exports the full dataset to `Coating_Results.xlsx`.



Output file structure

All data are saved automatically to `Coating_Results` on the Desktop:

```
Coating_Results/
  mappings/      # blank and post-coating .npz height maps
  dry_results/   # Coating_Results.xlsx
```

Each row in the Excel file is one measurement point. The exported columns are listed in the following table:

Column	Description
`sample_name`	User-defined sample identifier
`coating_mode`	UNIFORM, ZONE, or GRADIENT
`x_mm`	X-position of measurement point in mm
`y_mm`	Y-position of measurement point in mm
`z_blank_mm`	Pre-coating surface height in mm
`z_dry_mm`	Post-coating surface height in mm
`z_actual_gap_mm`	Effective gap at this point in mm
`thickness_mm`	Dry thickness in mm
`thickness_um`	Dry thickness in µm
`zone_index`	Zone assignment (1 or 2)
`set_speed_mm_s`	Set speed by user
`set_height_um`	Set height by user

Commissioning 1 — Bed Levelling

1. In the GUI, open the *Quick Bed Check* tab and click *Generate Heatmap*.
2. If the peak-to-valley deviation exceeds 200 µm, turn the spring-loaded corner knobs to correct the tilt and scan again. Repeat until the surface is sufficiently flat.

This step only needs to be repeated if the vacuum plate has been removed and reinstalled.

Commissioning 2 — Z-Offset Calibration

The Z-offset is the distance between the CR Touch trigger point and the blade tip. Set it once before first use; recalibrate only if the blade holder is mechanically modified.

1. Home the printer without a blade attached to avoid a crash with the blade storage. To do that navigate to *Control* → *Motion* on the printer display. The printer homes all axes automatically.
2. Attach a blade manually to the blade holder.
3. Navigate to *Move Z* and slowly decrease the Z height till $Z = 0$. At this position the blade should not yet touch the vacuum plate.
4. Navigate to *Prepare* → *Z-offset* and adjust the value until the blade just contacts the plate surface at $Z = 0$.
   - Blade does not reach the plate → increase the value
   - Blade presses too firmly → decrease the value
5. Save the Z-offset. The value is retained after power-off.

Commissioning 3 — Functional Verification

1. **Mapping —** In the GUI navigate to the *Mapping* tab, select Field 1 and click *Start Mapping*. The system maps the empty vacuum plate and saves the result. Check that a .npz file is generated in the folder /CoatingResults/mappings.
2. **Dry run —** Load one blade into slot 1. In the *Coating* tab, select Field 1, mode UNIFORM, gap 500 µm, speed 5 mm s^{-1}. The system picks up the blade, aligns the blade at the substrate surface and goes to its starting point. You will be asked to click OK and the blade will then traverse the full Y travel at 500 µm gap. At the end of the run the blade will be ejected. Verify that no collision occurs.

Completed without errors? The system is ready for electrode coating.

Safety Notes

Utility blades — Blades are sharp. Handle them carefully when loading the rack or collecting from the ejection tray. Wear cut-resistant gloves.

Moving axes — The motorised axes move without warning during automated sequences. Keep hands and objects clear of the motion envelope whenever the system is active.

Compressed air — Operating pressure is typically 4–6 bar. Check all pneumatic connections before each session and replace damaged tubing immediately.

Solvent compatibility — PLA is compatible with water-based slurry systems. For organic solvents verify compatibility or use PETG parts instead. See Section 2.